AKSHAT BAGLA

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Computational Mechanics | Structural Solver Development | Numerical Methods | GPU/HPC

HIGHLIGHTS

- Lead developer of **OpenSG**, an open-source multiscale structural solver in **Sandia's WETO toolkit**, used by **three universities**.
- Differentiable JAX solver: **25 million DOF on one A100 GPU**; on CPU, **5× faster than commercial SwiftComp**.
- **BAR-URC blade**: beam model within **1–2%** of a **2,000-CPU, 1–2 day 3D FEA**; full-blade **local buckling in 1.25 hr**.

EDUCATION

Ph.D. in Aerospace Engineering - Purdue University, West Lafayette, IN — GPA 3.78/4.00 | May 2027 (Expected)
Dissertation: *Differentiable Multiscale Constitutive Modeling of Thin-Walled Composite Beams Using Mechanics of Structure Genome*
Coursework: Elasticity, Structural Dynamics & Stability, Aeroelasticity, Scientific Computing

M.Tech. in Aerospace Engineering - Indian Institute of Technology Kanpur, India — CPI 9.5/10 | 2021

B.Tech. in Mechanical Engineering - G. B. Pant University, Pantnagar, India — 2018

TECHNICAL SKILLS

Formulation	Continuum mechanics, beam/plate/shell theories, variational asymptotics, geometric nonlinearity, dynamics
Numerics	FEM (C^0 Lagrange, C^1 Hermite), DG interior penalty, locking-free elements, periodic BCs, Lagrange multipliers
Solvers	Matrix-free CG, preconditioners (Jacobi, ILU, Chebyshev), JetSCI (JAX–PETSc), Pardiso, SLEPc eigensolvers
Software	Python, C++, Fortran, JAX (XLA/JIT, autodiff, vmap), FEniCSx/PETSc, MPI, gmsh, Cubit
Validation	Convergence studies, analytical, Abaqus, ANSYS, VABS, SwiftComp, Sierra

WORK EXPERIENCE

Lead Graduate Researcher (Purdue) - Cloud-Based Multiscale Modeling Toolkit — **Oct 2025 – Present**
NSF project with The University of Texas at Arlington and AFRL — West Lafayette, IN

- For GPUs, built OpenSG 2.0 in JAX on AFRL's *fea-in-jax* - matrix-free autodiff JVPs, preconditioned CG, rigid-body motion fixed.
- To get beam stiffness from shell meshes, derived Reissner–Mindlin and Kirchhoff–Love beam theories with C^1 Hermite and shear-locking-free quadrilateral elements - within 1% of 2D solid FE on thin walls in a thin-to-thick convergence study.
- To replace 3D FEA of lattice plates, derived a Reissner–Mindlin plate model from 3D elasticity (no shear correction factors) - in-plane stresses within 5%, frequencies within 6.1% (static, transient, modal), 25–320× cheaper.
- Traced inflated shear stiffness in Kirchhoff–Love beam properties to a junction artifact, not physics, and fixed it.
- To homogenize 3D shell lattices (TPMS), derived a shell-based model - within 5% of solid-element results, 4× faster.

AI Agent Developer - CompositesAI Platform — **May 2026 – Present**
Purdue University, with AnalySwift LLC (VABS) — West Lafayette, IN

- To run OpenSG through AI agents, built mesh-builder, homogenization and stress-recovery skills for shells and solids.
- Integrated CompositesAI with Claude Code, running CompositesAI and Abaqus in sync for structural analysis.
- Modeled the 138 m IEA 22 MW blade (windIO → blade geometry → shell sections), ∼1 min/station; 3D stresses match VABS.

Lead Graduate Researcher - Structural Design Tool: Wind Turbine Blade — **Jan 2023 – Sep 2025**
Sandia National Laboratories / U.S. DOE funded project in collaboration with NREL, Purdue University — West Lafayette, IN

- For an open-source design tool, built OpenSG 1.0 in FEniCSx/PETSc from weak forms - within 0.4% of SwiftComp and VABS.
- Derived a shell-element beam theory for curved, tapered sections - 10× faster and more accurate than PreComp and OptiStruct.
- Since tapered blades are not periodic, derived boundary-solution constraints that yield Timoshenko and mass matrices.
- To predict local buckling without full 3D FEA, used the beam model's pre-stress and recovered 3D stress in a geometrically nonlinear (Green–Lagrange) eigenproblem via Krylov–Schur - within 5% of analytical and Sierra solutions.

Structural Design Engineer - Aerial Vehicles — **Oct 2021 – Jul 2022**
EndureAir Systems Pvt. Ltd. — India

- For tail-sitter and helicopter UAVs, built FE models and led structural design - modal analysis drove design changes.

Research Assistant (M.Tech.) - Indian Institute of Technology Kanpur — **Aug 2019 – Aug 2021**
Thesis: *A Continuum Damage Mechanics-Based Model for Chopped Prepreg-Based Discontinuous Composites* — India

- Improved the RSA algorithm with a periodic mesh - chip packing from 32% to 59%; automated per-chip orientation in Abaqus.
- For strength and progressive damage, built a homogenization code and a linear CDM Abaqus UMAT - validated by experiments.

JOURNAL MANUSCRIPTS

- K. Bonney, **A. Bagla**, E. Camarena, W. Yu, *OpenSG: A FEniCSx-Based Implementation of the Mechanics of Structure Gene with Emphasis on Aperiodic Beams*, Journal of Open Source Software (JOSS), under review, 2026.
- **A. Bagla**, W. Yu, *Three-Dimensional Stress Recovery from a Reissner-Like Model for Lattice Plates Based on Mechanics of Structure Genome*, manuscript prepared for submission, 2026.
- P. R. Krishna Mohan, **A. Bagla**, P. M. Mohite, *Development, Characterisation and Strength Prediction of Chopped Prepreg-Based Discontinuous Composites*, manuscript prepared for submission, 2026.

BOOK CHAPTERS

- **A. Bagla**, M. K. Ballard, W. Yu, *OpenSG: A Native JAX-MSG Framework for Thin-Walled Composite Beam*, in American Society for Composites (ASC) 41st Technical Conference, Springer, 2026.
- **A. Bagla**, E. Camarena, W. Yu, *Theory of Arbitrary Shaped Thin-Walled Composite Beams*, in Proceedings of the American Society for Composites 40th Technical Conference, Vol. II, 2026.
- **A. Bagla**, E. Camarena, W. Yu, *Three-Dimensional Structural Analysis of Shell Structures Using MSG-Based Beam Model*, in Emerging Composites and Applications, Springer, pp. 163–177, 2026.
- **A. Bagla**, P. R. Krishna Mohan, P. M. Mohite, *Damage Initiation and Strength Predictions of Randomly Oriented Strands of Prepreg-Based Discontinuous Composites*, in Structural Integrity Conference and Exhibition (SICE), Springer Nature, pp. 275–286, 2022.
- S. Borchate, **A. Bagla**, P. M. Mohite, *A CDM-Based Modelling and Prediction of Progressive Damage in Z-Pinned Unidirectional Laminated Composites*, in Structural Integrity Conference and Exhibition (SICE), Springer Nature, pp. 365–377, 2022.

TECHNICAL REPORT

- **A. Bagla**, E. Camarena, W. Yu, *OpenSG: A Multiscale Constitutive Modeling Approach for Realistic Thin-Walled Composite Structures*, Sandia National Laboratories Report, OSTI, 2025.

CONFERENCE PAPERS

- **A. Bagla**, M. K. Ballard, W. Yu, *Multiscale Modeling of Composites with AI/ML Framework Using JAX*, ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM), Long Beach, CA, June 2026.
- **A. Bagla**, E. Camarena, W. Yu, *Timoshenko Beam Modeling of Realistic Wind Turbine Blades Using Mechanics of Structure Genome*, ASME International Mechanical Engineering Congress and Exposition (IMECE), Memphis, TN, 2025.
- **A. Bagla**, W. Yu, E. Camarena, *OpenSG: An Open-Source Multiscale Constitutive Modeling Code for Composite Structures and Materials*, American Society for Composites (ASC) 38th Technical Conference, Boston, MA, 2023.

MANUSCRIPTS IN PREPARATION

- **A. Bagla** et al., *Scalable, Differentiable Constitutive Modeling of Composite Structures Using Mechanics of Structure Genome*.
- **A. Bagla** et al., *Timoshenko Beam Modeling of Thin-Walled Composite Beams Using Reissner–Mindlin Shell-Based MSG Theory*.
- **A. Bagla** et al., *A Continuous/Discontinuous Galerkin Interior-Penalty Kirchhoff Shell Model for Thin-Walled Beam Properties*.

CONFERENCE PRESENTATIONS

- ASME SSDM 2023, 2024, 2026 | ASME IMECE 2025 | ASC Technical Conference 2023–2026 | SES 2026

TEACHING & AWARDS

- **Graduate TA, Purdue** (2023–26) - Aircraft Structural Analysis, Mechanics of Composite Materials, Aeromechanics II.
- NSF Travel Award ($1,000) | Purdue Recognition Award ($1,500) | Top 5% of cohort, IIT Kanpur

REFERENCES

Prof. Wenbin Yu - Ph.D. advisor
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